Class 11 Chemistry Important Questions

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Unit 1: Some Basic Concepts of Chemistry

KEY FORMULAE QUICK RECAP

- **Mole (n)** = Mass / Molar Mass = N / N_A = Volume of gas at STP (L) / 22.4 L
- **Molarity (M)** = Moles of Solute / Volume of Solution (L)
- **Molality (m)** = Moles of Solute / Mass of Solvent (kg)

Q1. Define Limiting Reagent with a suitable example. [2 Marks]

Answer: The limiting reagent is the reactant that is completely consumed first in a chemical reaction and limits the amount of product formed.
Example: In the reaction $2H_2 + O_2 \rightarrow 2H_2O$, if 2 moles of H_2 react with 2 moles of O_2, hydrogen is the limiting reagent because 2 moles of H_2 require only 1 mole of O_2.

Q2. A compound contains 4.07% hydrogen, 24.27% carbon, and 71.65% chlorine. Its molar mass is 98.96 g/mol. Determine its empirical and molecular formulas. [3 Marks]

Solution Steps:

1. **Relative moles:** C = 24.27/12 = 2.02, H = 4.07/1 = 4.07, Cl = 71.65/35.5 = 2.02
2. **Simplest mole ratio:** Divide by smallest (2.02) → C = 1, H = 2, Cl = 1
3. **Empirical Formula:** CH_2Cl (Empirical formula mass = 12 + 2 + 35.5 = 49.5 g/mol)
4. **n factor:** Molar Mass / Empirical Mass = 98.96 / 49.5 ≈ 2
5. **Molecular Formula:** $(CH_2Cl)_2 = C_2H_4Cl_2$

Q3. Calculate the molarity of a solution prepared by dissolving 4 g of NaOH in enough water to form 250 mL of solution. [2 Marks]

Solution:
Molar mass of NaOH = 23 + 16 + 1 = 40 g/mol.
Moles of NaOH = 4 / 40 = 0.1 mol.
Volume of solution = 250 mL = 0.250 L.
Molarity (M) = 0.1 / 0.250 = **0.4 M**.

Unit 2: Structure of Atom

Q4. State and explain Heisenberg's Uncertainty Principle. Give its mathematical expression. [2 Marks]

> **Answer:** It states that it is impossible to determine simultaneously both the exact position and exact momentum (or velocity) of a microscopic particle like an electron.
> **Mathematical Expression:** $\Delta x \times \Delta p \geq h / (4\pi)$ or $\Delta x \times (m \Delta v) \geq h / (4\pi)$,
> where Δx is position uncertainty, Δp is momentum uncertainty, and h is Planck's constant.

Q5. Calculate the wavelength of a photon emitted during a transition from $n = 5$ state to the $n = 2$ state in a hydrogen atom. [3 Marks]

> **Solution:**
> Using Rydberg Formula: $1 / \lambda = R_H (1/n_1^2 - 1/n_2^2)$ where $R_H = 1.097 \times 10^7\ m^{-1}$
> $1 / \lambda = 1.097 \times 10^7 (1/2^2 - 1/5^2) = 1.097 \times 10^7 (1/4 - 1/25) = 1.097 \times 10^7 (21/100)$
> $1 / \lambda = 2.3037 \times 10^6\ m^{-1} \Rightarrow \lambda = 4.34 \times 10^{-7}\ m = 434\ nm$ (Visible Balmer series, Blue light).

Q6. Explain Pauli's Exclusion Principle and Hund's Rule of Maximum Multiplicity. [3 Marks]

> • **Pauli's Exclusion Principle:** No two electrons in an atom can have the same set of all four quantum numbers (n, l, m_l, m_s). An orbital can hold a maximum of 2 electrons with opposite spins.
> • **Hund's Rule:** Electron pairing in orbitals of the same subshell (degenerate orbitals) does not take place until each orbital is singly occupied with parallel spins.

Unit 3: Classification of Elements & Periodicity in Properties

Q7. Why is the first ionization enthalpy of Nitrogen higher than that of Oxygen, despite Oxygen having a higher atomic number? [2 Marks]

> **Answer:** Nitrogen has electronic configuration $1s^2\ 2s^2\ 2p^3$, which features a *half-filled 2p subshell* (extra stable). Oxygen has $1s^2\ 2s^2\ 2p^4$. Removing an electron from Nitrogen requires extra energy to break this stable half-filled configuration, whereas removing an electron from Oxygen results in a stable half-filled $2p^3$ configuration.

Q8. Explain why electron gain enthalpy of Chlorine is more negative than that of Fluorine. [2 Marks]

> **Answer:** Fluorine's $2p$ subshell is very compact in size. Adding an extra electron to Fluorine causes strong inter-electronic repulsions in the small $2p$ subshell. Chlorine has a larger $3p$ subshell, so the incoming electron experiences less repulsion, releasing more energy (more negative $\Delta_{eg}H$).

Unit 4: Chemical Bonding & Molecular Structure

Q9. On the basis of VSEPR theory, predict the shape and geometry of: (i) SF_4, (ii) XeF_4, (iii) NH_3.

[3 Marks]

Molecule	Bond Pairs + Lone Pairs	Geometry	Shape
SF_4	4 BP + 1 LP (Steric No. 5)	Trigonal Bipyramidal	See-Saw
XeF_4	4 BP + 2 LP (Steric No. 6)	Octahedral	Square Planar
NH_3	3 BP + 1 LP (Steric No. 4)	Tetrahedral	Trigonal Pyramidal

Q10. Define Hybridization. Explain the hybridization and structure of PCl_5 and why its axial bonds are longer than equatorial bonds.

[3 Marks]

Definition: Hybridization is the intermixing of atomic orbitals of slightly different energies to form an equal number of new hybrid orbitals of equivalent energy and shape.
PCl_5 Hybridization: sp^3d (Trigonal Bipyramidal geometry).
Bond Length Difference: The two axial bonds experience greater repulsion from three equatorial bonds at 90° compared to equatorial-equatorial repulsions at 120°. To minimize repulsion, axial bonds elongate and become weaker.

Q11. Draw the Molecular Orbital (MO) energy diagram for O_2 molecule. Calculate its bond order and explain its paramagnetic behavior.

[5 Marks]

Electronic Configuration of O_2 (16 e^-):
$\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ (\pi 2p_x^2 = \pi 2p_y^2)\ (\pi^*2p_x^1 = \pi^*2p_y^1)$
Bond Order: $½\ (N_b - N_a) = ½\ (10 - 6) = 2$ (Double Bond).
Paramagnetism: Presence of 2 unpaired electrons in degenerate antibonding orbitals (π^*2p_x, π^*2p_y).

Unit 5: Chemical Thermodynamics

Q12. State Hess's Law of Constant Heat Summation with an application.

[2 Marks]

Statement: The total enthalpy change during a chemical reaction remains the same whether the reaction takes place in a single step or in a series of steps.
Application: Used to determine enthalpies of reactions that cannot be measured directly in experiments (e.g., formation of CO from C and O_2).

Q13. Derive the condition for spontaneity of a reaction in terms of Gibbs Energy (ΔG).

[3 Marks]

Gibbs-Helmholtz Equation: $\Delta G = \Delta H - T\Delta S$
- If $\Delta G < 0$ **(Negative)**: Reaction is *Spontaneous*.
- If $\Delta G > 0$ **(Positive)**: Reaction is *Non-spontaneous*.
- If $\Delta G = 0$: System is in *Equilibrium*.

Unit 6: Chemical Equilibrium

Q14. State Le Chatelier's Principle. Predict the effect of (a) increasing pressure and (b) increasing temperature on the synthesis of Ammonia: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \Delta H = -92.4\ kJ$. [3 Marks]

Principle: If a system at equilibrium is subjected to a change in temperature, pressure, or concentration, the system shifts in a direction that tends to counteract the effect of the change.
• **Increasing Pressure:** Shifts equilibrium in the forward direction (towards fewer moles of gas, 4 mol → 2 mol).
• **Increasing Temperature:** As reaction is exothermic ($\Delta H < 0$), raising temperature shifts equilibrium in the backward direction (lowers yield of NH_3).

Q15. Define Buffer Solution and Buffer Capacity. Give one example each of acidic and basic buffer. [2 Marks]

Buffer Solution: A solution that resists changes in pH upon addition of small amounts of an acid or a base.
• **Acidic Buffer:** $CH_3COOH + CH_3COONa$ (Weak acid + its salt with strong base).
• **Basic Buffer:** $NH_4OH + NH_4Cl$ (Weak base + its salt with strong acid).

Unit 7: Redox Reactions

Q16. Balance the following redox reaction in acidic medium by Ion-Electron Method:
$MnO_4^-(aq) + Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + Fe^{3+}(aq)$ [3 Marks]

Step-by-step Solution:
1. **Oxidation Half Reaction:** $Fe^{2+} \rightarrow Fe^{3+} + e^-$ × 5
2. **Reduction Half Reaction:** $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$
3. **Balanced Overall Ionic Equation:**
$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

Unit 8: Organic Chemistry – Basic Principles & Techniques

Q17. Explain Inductive Effect, Electromeric Effect, Resonance Effect, and Hyperconjugation with suitable examples. [4 Marks]

• **Inductive Effect (+I / -I):** Permanent displacement of σ-electrons along a carbon chain due to electronegativity difference (e.g., -Cl in CH_3CH_2Cl pulls σ-electrons).
• **Resonance Effect:** Delocalization of π-electrons in conjugated systems giving extra stability (e.g., Benzene).
• **Hyperconjugation (No-bond resonance):** Delocalization of σ-electrons of C-H bond of an alkyl group directly attached to an unsaturated atom or carbocation. Explains stability order of carbocations: 3° > 2° > 1°.

Q18. Write IUPAC names for the following structures: [3 Marks]

1. $CH_3\text{-}CH(OH)\text{-}CH_2\text{-}COOH$ → **3-Hydroxybutanoic acid**
2. $CH_3\text{-}C(=O)\text{-}CH_2\text{-}CH=CH_2$ → **Pent-4-en-2-one**
3. $(CH_3)_3C\text{-}CH_2\text{-}CH_3$ → **2,2-Dimethylbutane**

Unit 9: Hydrocarbons

Q19. State Markownikoff's Rule and Anti-Markownikoff's Rule (Peroxide Effect) with mechanisms and examples. [4 Marks]

• **Markownikoff's Rule:** When an unsymmetrical reagent (e.g., HBr) adds to an unsymmetrical alkene, the negative part adds to the carbon having fewer hydrogen atoms.
Example: $CH_3\text{-}CH=CH_2 + HBr \rightarrow CH_3\text{-}CH(Br)\text{-}CH_3$ (2-Bromopropane - Major product via stable 2° carbocation).
• **Anti-Markownikoff's Rule:** In the presence of peroxide, HBr adds opposite to Markownikoff's rule via a free radical mechanism.
Example: $CH_3\text{-}CH=CH_2 + HBr \rightarrow CH_3\text{-}CH_2\text{-}CH_2Br$ (1-Bromopropane).

Q20. Explain the electrophilic substitution mechanism of Nitration and Friedel-Crafts Alkylation of Benzene. [4 Marks]

• **Nitration:** Benzene reacts with conc. HNO_3 + conc. H_2SO_4 at 323K to yield Nitrobenzene. Electrophile generated: NO_2^+ (Nitronium ion).
• **Friedel-Crafts Alkylation:** Benzene reacts with CH_3Cl in presence of anhydrous $AlCl_3$ to give Toluene. Electrophile generated: CH_3^+.

💡 **Final Revision Tip:** Practice drawing chemical structures, VSEPR shapes, and balancing redox reactions by hand. Focus heavily on IUPAC rules and organic mechanisms for maximum scoring in board exams.